

Shubhradeep Roy

Kolkata, India | shubhradeeproxy343@gmail.com | [linkedin.com/in/shubhradeep-roy](https://www.linkedin.com/in/shubhradeep-roy) | github.com/Esar1212
Full Stack Developer (Next.js — React — Node.js)

EDUCATION

Heritage Institute of Technology

Bachelor of Technology in Computer Science and Engineering, CGPA: 9.64/10

Kolkata, India

Sept. 2023 – Sept. 2027

TECHNICAL SKILLS

Languages: JavaScript, SQL, Python

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Socket.io

Technologies: MongoDB, WebSockets, Blockchain

PROJECTS

QuestEd - Online Examination Platform | *Next.js, MongoDB, Gemini AI, JWT, WebSockets, NextAuth*

- Built and deployed full-stack exam platform with real-time monitoring and AI-based grading
[Live App Link](#) — [GitHub Repo Link](#)
- Integrated Gemini 2.5 Flash model for intelligent question generation (MCQ & descriptive) and automatic evaluation of descriptive answers by semantic comparison with model answers, reducing grading time by 80%
- Built secure JWT-based authentication with Google OAuth
- Implemented anti-cheating mechanisms with tab-switch detection triggering automatic submission, ensuring exam integrity
- Designed scalable MongoDB architecture with role-based dashboards for teachers and students featuring real-time analytics

Decentralized Voting dApp | *Solidity, Ethereum, React, Vite, MetaMask, ethers.js, WebSockets*

- Developed blockchain-based voting application with Solidity smart contracts deployed on Sepolia TestNet ensuring transparent, tamper-proof elections with one-vote-per-wallet enforcement — [Live App Link](#) — [GitHub](#)
- Implemented secure access control using Solidity modifiers restricting candidate management and poll administration to contract owner
- Built responsive React frontend with MetaMask wallet integration for secure authentication and transaction signing via ethers.js
- Engineered real-time voting analytics dashboard using WebSocket backend (deployed on Render) for live vote count updates

PROFESSIONAL EXPERIENCE

MERN Development Intern

Sept. 2024 – Jan. 2025

ARDENT COMPUTECH PVT LTD

Remote

- Architected full-stack Online Examination Platform using MERN stack serving 500+ concurrent users with real-time exam management and automated grading workflows
- Engineered WebSocket-based real-time system for live monitoring and synchronized states across distributed clients
- Enhanced system performance by 40% through MongoDB query optimization and asynchronous processing

Data Science Intern

Oct. 2023 – Dec. 2023

Elenxia

Remote

- Developed data analysis pipelines using NumPy and Pandas; created visualizations with Matplotlib and Seaborn
- Implemented web scraping with BeautifulSoup and applied inferential statistics (Z-Test, F-Test) for data analysis

CERTIFICATIONS & ACHIEVEMENTS

Hackathons: HackHeritage 3.0 participant | **Certified:** Full Stack MERN Development, Data Science with Python